

# Şirag Erkol

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| CONTACT<br>INFORMATION | Northwestern University<br>Center for Science of Science and Innovation<br>1800 Sherman Ave, Evanston, IL 60201                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | (812) 558-8938<br>siragerkol@gmail.com<br>siragerkol.github.io                                                                   |
| ACADEMIC<br>POSITIONS  | <b>Northwestern University</b> , Evanston, IL<br><i>Research Assistant Professor</i><br><b>Northwestern University</b> , Evanston, IL<br><i>Postdoctoral Researcher</i><br><b>Indiana University</b> , Bloomington, IN<br><i>Research &amp; Teaching Assistant</i><br><b>Boğaziçi University</b> , Istanbul, Turkey<br><i>Research &amp; Teaching Assistant</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>July 2025 – present</b><br><b>July 2023 – June 2025</b><br><b>August 2017 – May 2023</b><br><b>October 2014 – August 2017</b> |
| EDUCATION              | <b>Indiana University</b> , Bloomington, IN<br><i>Ph.D. in Informatics, Complex Networks and Systems track</i> <ul style="list-style-type: none"><li>• Advisor: <i>Assoc. Prof. Filippo Radicchi</i></li><li>• Thesis: <i>Identifying Influential Spreaders in Complex Networks</i></li><li>• Minor: Computer Science</li></ul> <i>M.S. in Informatics</i><br><b>Boğaziçi University</b> , Istanbul, Turkey<br><i>M.S. in Industrial Engineering</i> <ul style="list-style-type: none"><li>• Advisor: <i>Assoc. Prof. Gönenç Yücel</i></li><li>• Thesis: <i>Influence Maximization Based on Partial Network Structure Information: A Comparative Analysis on Seed Selection Heuristics</i></li></ul> <i>B.S. in Industrial Engineering</i> <ul style="list-style-type: none"><li>• Graduation Project: <i>Analyzing Parking Stand Requirements for Istanbul Ataturk Airport</i></li></ul> | <b>August 2017 – May 2023</b><br><b>September 2014 – August 2016</b><br><b>September 2010 – June 2014</b>                        |
| WORK<br>EXPERIENCE     | <b>Pacific Northwest National Laboratory</b> , Richland, WA<br><i>Ph.D. Intern</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>June 2021 – August 2021</b>                                                                                                   |
| HONORS &<br>AWARDS     | IU Luddy Summer Research Award (\$4400)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>2022</b>                                                                                                                      |
| PUBLICATIONS           | <b>Published:</b><br>Symmetry breaking in optimal transport networks<br>S. Patwardhan, M. Barthelemy, <b>Ş. Erkol</b> , S. Fortunato, F. Radicchi, <i>Nature Communications</i> , 15, 3758 (2024)<br>[DOI] [arXiv]<br>Consistency pays off in science<br><b>Ş. Erkol</b> , S. Sikdar, F. Radicchi, and S. Fortunato, <i>Quantitative Science Studies</i> , 4, 491 (2023)<br>[DOI] [arXiv] [GitHub] [Website]<br>Effective submodularity of influence maximization on temporal networks<br><b>Ş. Erkol</b> , D. Mazzilli and F. Radicchi, <i>Physical Review E</i> , 106, 034301 (2022)<br>[DOI] [arXiv]                                                                                                                                                                                                                                                                                   |                                                                                                                                  |

Who is the best coach of all time? A network-based assessment of the career performance of professional sports coaches

**Ş. Erkol** and F. Radicchi, *Journal of Complex Networks*, 9, cnab012 (2021)

[DOI] [arXiv] [Website]

Influence maximization on temporal networks

**Ş. Erkol**, D. Mazzilli and F. Radicchi, *Physical Review E*, 102, 042307 (2020)

[DOI] [arXiv]

Systematic comparison between methods for the detection of influential spreaders in complex networks

**Ş. Erkol**, C. Castellano and F. Radicchi, *Scientific Reports*, 9, 15095 (2019)

[DOI] [arXiv]

Influence maximization in noisy networks

**Ş. Erkol**, A. Fageeh and F. Radicchi, *EPL*, 123, 58007 (2018)

[DOI] [arXiv]

Influence Maximization Based on Partial Network Structure Information: A Comparative Analysis on Seed Selection Heuristics

**Ş. Erkol** and G. Yücel, *International Journal of Modern Physics C*, 28, 1750122 (2017)

[DOI]

#### PRESENTATIONS

##### Talks:

Effective submodularity of influence maximization on temporal networks,  
NetSci 2022, Online

**July 2022**

Influence maximization on temporal networks, Networks2021, Online

**July 2021**

Influence maximization on temporal networks, University of Limerick,  
Ireland, Online

**March 2021**

##### Posters:

Consistency pays off in science  
ICSSI 2023, Evanston, IL, USA

**June 2023**

Systematic comparison between methods for the detection of influential  
spreaders in complex networks  
NetSci 2020, Online

**September 2020**

Influence maximization in noisy networks  
NetSci 2019, Burlington, VT, USA

**May 2019**

#### COMMUNITY SERVICE

**Journal Reviewer:** Applied Mathematics and Computation; Chaos, Solitons & Fractals; EPJ Data Science; Expert Systems with Applications; IEEE Transactions on Network Science and Engineering; Journal of Complex Networks; Journal of Computational Science; Pattern Recognition; Physical Review E; Royal Society Open Science; Science Advances

**Conference Reviewer:** ISSI 2023; ICSSI 2024

**Journal Subreviewer:** PLoS One, PNAS

#### COMPUTER SKILLS

Frequent user of Python, L<sup>A</sup>T<sub>E</sub>X  
Experience in Django, NetLogo, HTML, CSS  
Familiar with MySQL, C, R, Git, Stella

#### LANGUAGES

English (fluent), Turkish (native), Armenian (native), French (beginner), Spanish (beginner)

